# Marco Ciccone


**Personal Information**

*Surname Name*: Ciccone Marco
*Nationality*: Italian
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**Research Interests**

Deep Learning, Meta-Learning, Continual Learning, Federated Learning, Transfer Learning, Computer Vision, Game Theory, Multi-Agent Learning

**Academic Experience**

**University College London**, UK

Honorary Research Fellow — **January 2022 - now**

**Politecnico di Torino**, Italy

Postdoctoral Researcher - ELLIS PostDoc Program — **June 2021 - now**
Advisor: Prof. Barbara Caputo (Politecnico di Torino)
Co-Advisor: Prof. Carlo Ciliberto (University College London)

**Work Experience**

**NVIDIA**, Santa Clara, California
*Research Intern* — **August 2018 - November 2018**

**NNAISENSE**, Lugano, Switzerland
*Research Intern* — **July 2017 - January 2018**

**Horus Technology**, Milan, Italy
*Machine Learning Engineer* — **June 2016 - June 2017**

**Projest S.p.A.**, Prato, Italy
*IT Consultant - Software Engineer* — **Nov 2009 - Oct 2012**

**Education**

**Politecnico di Milano**, Italy

PhD in Information Technology — **April 2021**

- Thesis: *"On Iterative and Conditional Computation for Visual Representation Learning"*
- Advisor: Prof. Matteo Matteucci
- Co-Advisor: PhD Jonathan Masci
- Grade: *cum Laude*

**Romanian Institute of Science and Technology**, Cluj-Napoca, Romania

Visiting PhD student, advised by PhD Luigi Malago' — **May-July 2018**

*"Information Geometry of Adversarial Robustness of Neural Networks"*

**Politecnico di Milano**, Italy

MSc in Computer Science and Engineering **April 2016**

- Thesis: *"Pixelwise Semantic Segmentation of urban scenes using Recurrent Deep Neural Networks"*
- Advisors: Prof. Matteo Matteucci, PhD Francesco Visin
- Language: English
- Grade: *110/110 cum Laude*
- GPA: *28.85/30*

**Università degli Studi di Firenze**, Italy

BSc in Computer Engineering **July 2013**

- Thesis: *"Blind Source Separation methods for multichannel audio signals based on Independent Component Analysis"*
- Advisor: Prof. Fabrizio Argenti
- Language: Italian
- Grade: 97/110

## Professional Activities

**Chair and Organizer**

- **Competition chair NeurIPS 2022**
  (co-chair with Gustavo Stolovitzky and Jake Albrecht)
- **Demonstration and Competition chair NeurIPS 2021**
  (co-chair with Barbara Caputo and Douwe Kiela)

**Reviewer**

- Neural Information Processing Systems (NeurIPS)
- International Conference on Learning Representations (ICLR)
- AAAI Conference on Artificial Intelligence (AAAI)
- International Journal of Computer Vision (IJCV)
- International Conference on Robotics and Automation (ICRA)
- International Conference on Intelligent Robots and Systems (IROS)
- Geometric meets Deep Learning Workshop (ICCV)
- Neural Computing and Applications Journal (NCAA)
- Geometric Deep Learning Workshop (ECCV)

## Attended Schools/Events

**Transylvanian Machine Learning Summer School 2018,** Cluj, Romania

*Acceptance Rate: 8%* **16-22 July 2018**

**Machine Learning Summer School 2017,** Tubingen, Germany

*Acceptance Rate: 13%* **19-30 June 2017**

**International Computer Vision Summer School 2017,** Sicily, Italy

*Acceptance Rate: 35%* **9-15 July 2017**

**Self-Organizing Conference on Machine learning 2018,** Toronto, Canada

*Selected on application* **30 Nov - 1 Dec 2018**

## Teaching

**Mediterranean Machine Learning School (M2L) 2022**

Lab Tutor

*Colab on Graph Neural Networks* - Tutorials 2022 **September 2022**

**Mediterranean Machine Learning School (M2L) 2021**

Lab Tutor

*Colab on Variational Auto-Encoders* - Tutorials 2021 **January 2021**

**Deep Learning 2017/2018**, Prof. Matteo Matteucci

PhD and MSc Course at Politecnico di Milano

*Teaching Assistant (PyTorch and Tensorflow Tutorials)* **February 2018**

**Cognitive Robotics 2016/2017**, Prof. Matteo Matteucci

MSc Course at Politecnico di Milano

*Teaching Assistant (Fundamentals of Deep learning)* **June 2017**

## Publications

[1] D. Shenaj*, E. Fanì*, M. Toldo, D. Caldarola, A. Tavera, U. Michieli†, **M. Ciccone**†, P. Zanuttigh†, and B. Caputo†, "Learning across domains and devices: Style-driven source-free domain adaptation in clustered federated learning," in *2023 IEEE Winter Conference on Applications of Computer Vision (WACV)*, IEEE, 2023

[2] D. Caldarola*, B. Caputo, and **M. Ciccone***, "Improving generalization in federated learning by seeking flat minima," in *European Conference on Computer Vision*, 2022

[3] L. Fantauzzo, E. Fani, D. Caldarola, A. Tavera, F. Cermelli, **M. Ciccone**, and B. Caputo, "Feddrive: Generalizing federated learning to semantic segmentation in autonomous driving," in *Proceedings of the 2022 IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2022

[4] R. Zaccone, A. Rizzardi, D. Caldarola, **M. Ciccone**, and B. Caputo, "Speeding up heterogeneous federated learning with sequentially trained superclients," in *26th International Conference on Pattern Recognition (ICPR)*, IEEE, 2022

[5] L. Carminati, F. Cacciamani, **M. Ciccone**, and N. Gatti, "A marriage between adversarial team games and 2-player games: Enabling abstractions, no-regret learning, and subgame solving," in *Proceedings of the 39th International Conference on Machine Learning*, Proceedings of Machine Learning Research, PMLR, 17–23 Jul 2022

[6] F. Cermelli*, D. Fontanel*, A. Tavera*, **M. Ciccone**, and B. Caputo, "Incremental learning in semantic segmentation from image labels," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2022

[7] L. Carminati, F. Cacciamani, **M. Ciccone**, and N. Gatti, "Public information representation for adversarial team games," in *NeurIPS 2021 Workshop Cooperative AI*, 2021 **Oral and Best paper award, CooperativeAI Workshop NeurIPS 2021**

[8] M. Planamente*, C. Plizzari*, M. Cannici*, **M. Ciccone**, F. Strada, A. Bottino, M. Matteucci, and B. Caputo, "Da4event: Towards bridging the sim-to-real gap for event cameras using domain adaptation," *IEEE Robotics and Automation Letters*, vol. 6, no. 4, pp. 6616–6623, 2021

[9] D. Caldarola, M. Mancini, F. Galasso, **M. Ciccone**, E. Rodola, and B. Caputo, "Cluster-driven graph federated learning over multiple domains," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, pp. 2749–2758, June 2021

[10] M. Cannici, C. Plizzari, M. Planamente, **M. Ciccone**, A. Bottino, B. Caputo, and M. Matteucci, "N-rod: A neuromorphic dataset for synthetic-to-real domain adaptation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, pp. 1342–1347, June 2021

[11] F. Cacciamani, A. Celli, **M. Ciccone**, and N. Gatti, "Multi-agent coordination in adversarial environments through signal mediated strategies," in *Proceedings of the 20th International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, 2021

[12] M. Cannici, **M. Ciccone**, A. Romanoni, and M. Matteucci, "A differentiable recurrent surface for asynchronous event-based data," in *Proceedings of the European Conference on Computer Vision (ECCV)*, pp. 1–17, Springer, 2020

[13] M. Cannici, **M. Ciccone**, A. Romanoni, and M. Matteucci, "Asynchronous convolutional networks for object detection in neuromorphic cameras," *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2019
**Best paper award, Second International Workshop on Event-based Vision and Smart Cameras Workshop at CVPR2019**

[14] M. Cannici, **M. Ciccone**, A. Romanoni, and M. Matteucci, "Attention mechanisms for object recognition with event-based cameras," *Proceedings of the IEEE Winter Conference on Applications of Computer Vision (WACV)*, 2019

[15] **M. Ciccone***, M. Gallieri*, J. Masci, C. Osendorfer, and F. Gomez, "Nais-net: Stable deep networks from non-autonomous differential equations," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2018

[16] F. Lattari*, **M. Ciccone***, M. Matteucci, J. Masci, and F. Visin, "Reconvnet: Video object segmentation with spatio-temporal features modulation," *The 2018 DAVIS Challenge on Video Object Segmentation - CVPR Workshops*, 2018

[17] A. Romanoni, **M. Ciccone**, F. Visin, and M. Matteucci, "Multi-view stereo with single-view semantic mesh refinement," in *Proceedings of the IEEE International Conference on Computer Vision Workshops (ICCVW)*, pp. 706–715, 2017

[18] F. Visin, **M. Ciccone**, A. Romero, K. Kastner, C. Kyunghyun, Y. Bengio, M. Matteucci, and A. Courville, "ReSeg : A Recurrent Neural Network-based Model for Semantic Segmentation," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2016
**Best paper award, DeepVision Workshop at CVPR2016**